1 A student wishes to measure the resistivity of glass. A teacher suggests that its resistivity is of the order of $10^6\,\Omega\,\text{m}$ which is very large.

For Examiner's Use

Resistivity ρ is defined by the equation

$$\rho = \frac{RA}{l}$$

where R is resistance, A is cross-sectional area and l is the length of the material.

The student is given a number of sheets of glass of the same thickness and of different areas.

Design a laboratory experiment to determine the resistivity of glass. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) how the glass would be connected to the circuit,

(c) the measurements that would be taken,

(d) the control of variables,

(e) how the data would be analysed,

(f) any safety precautions that you would take. [15]

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